

Sultan Of Oman
Ministry of Health
Royal Hospital



سلطنة عمان
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المستشفى السلطاني

Document Management System

	Sultanate of Oman Ministry of Health The Royal Hospital Department of Blood Transfusion Services
Code: BTS-GUD-1-Vers.6	
Effective Date: 24/09/2023	
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Title: Administration of blood and blood components	
1. Introduction: Blood transfusion is the process of transfusing blood or blood components into one's circulation intravenously. Transfusion is used for various medical conditions to replace lost components of bloods. The transfusion of Blood and Blood components carry a significant risk if not performed according to national guidelines (MOH, The Directorate General for Nursing Affairs, Blood and Blood Product Transfusion Policy). 1.1 Purpose: To guide nurses on how to prepare a patient for transfusion of blood components. Ensure that procedures of correct identification of patients during blood sampling and blood component administration have been strictly followed and adhered to in order to prevent potentially fatal errors. 1. Equipment: • Trolley/Tray washed with soap & water & cleaned with 70% spirit. • Intravenous cannula(14G, 16G or 18 G) • Sterile Normal Saline 0.9% Infusion Pack. • Blood component set with appropriate filter (except for granulocyte infusion where no filter should be used). • I.V. drip Stand/ Infusion pump. • Bandage and splint (Only if cannula is sited near the joint or if patient is restless). • Pressure Bag and Blood Warmer (In-Case of Emergency) Policy and Procedure: • Pre Blood Transfusion : Prior to ordering red blood cells for a patient, a sample of blood must be taken from the patient to check blood group and screening for antibodies, to ensure appropriate blood is transfused. This sample is valid for 72 hours to 7 days prior to transfusion depending on when was the last red cell transfusion. Patient with a transfusion history of less than 72 hours will require a new sample 24 hours prior to the planned transfusion, while patient with a transfusion history of 4 weeks or more; the new sample will be valid for 7 days. 1. Pre transfusion blood sampling: • All orders for group and screen/ cross match on in patients are electronically ordered by the doctor. Following the placement of the order a blood bank request form is generated. • Pre transfusion blood sampling (cross match) should only be undertaken by the staff that is trained and competency assessed. • The collection of a minimum of 4ml blood in EDTA tube is needed for cross match. • For paediatric patient (Age 2-5 years) sample should not be less than 2ml of blood in an EDTA tube. • For patients with no historical record of blood grouping at the Royal hospital, the nurse must collect two samples at two different occasions with an interval of half an hour or more. • For any further Investigation (e.g. antibody investigation) blood bank will request the ward for more blood sample. • State the urgency of the request clearly on the request form. When an urgent sample is sent, please contact the blood bank and sample should be label as urgent sample. • sample tubes must be labeled at the patient side by the individual who took the sample containers MUST NOT be labeled prior to the collection of the sample. Never ask another person to label a sample for you. • If the patient is conscious, verbally ask to give their full name, date of birth (their blood group). These details are checked against the patient identification band by using the 3V. • If patient is unconscious or unable to state their name, the patients' identification band must be checked against written documentation.	

- Samples should be transported to and received by the blood bank in a timely manner.
- For emergency or urgent samples, the hospital transfusion laboratory (blood bank) staff should be contacted to alert them to imminent receipt of the sample
- Staff member transporting urgent samples to the laboratory should be aware of the urgency of the situation and know where / who in the transfusion laboratory they should deliver it to.

C. Consent:

Blood and blood component transfusion is a treatment modality with potentially serious adverse patient outcomes, there is an ethical obligation to obtain patient consent to a blood transfusion wherever possible. The indication for the blood transfusion should be discussed between the clinician and the patient. A formal written consent is required to be filled then scanned to be saved in Al Shifa system.

The due attention given to the risk / benefit analysis and whether there are any alternatives. This discussion gives the patient a clear and contextual understanding of the risks of transfusion (the risks will vary depending upon the patient s condition). A record of this discussion must be documented in the patient notes.

Note:

- Valid consent for blood transfusion should be obtain and documented in the patient's clinical record by the Health care professional.
- For Non-chronically transfused Patient, consent is renewed upon each patient admission.
- Patient on long-term blood support (e.g. Patients with underlying haemoglobinopathies and those undergoing chemotherapy) should provide consent annually with an option to withdraw their consent any time during that period.
- In emergency situation where the patient or their guardian is unable or unavailable to provide consent, Pre-transfusion consent requirement are waived, in these cases, patients or guardians are retroactively informed about the requirement of and the reason of performing blood transfusion.
- In- Patients who are unable to obtain the consent by themselves (e.g. children, unconscious or confused patients, if there is language barrier), consent should be obtaining from a parent or caregiver.

D. Environment for Blood component Transfusion:

Transfusions must only be given in clinical areas where:

- Patients can be readily observed for possible reactions
- Resuscitation facilities are available including an AED and anaphylaxis treatment.
- Blood for more than one patient awaiting transfusion, for example in a treatment room or at the nurses' station, is extremely hazardous, as one of them may be ABO incompatible with your patient. This must be avoided by arranging for collection and commencement of the transfusion of blood components one patient at a time.

G. Blood component from Blood Bank:

Health care worker may only collect blood components if they have been trained

- When collecting the blood from a blood bank, the patient details must be carefully checked visually in addition to bar code scanning (if available) between the compatibility label, the blood component and the pickup slip to ensure that details are matched identically.
- Blood components must be collected for one patient at a time in all circumstances.
- One unit of red cells should be collected at a time unless extremely rapid transfusion of large quantities of blood is needed.
- The blood should be delivered to the relevant clinical area without delay. The clinical staff accepting the blood must check that the blood received is for the correct patient. The time of removal of the blood from the blood fridge is recorded in Blood slip records.
- Clinical staff must check the blood component on arrival to a clinical area using the blood slip with al-Shifa system.
- Blood not used within half an hour on the ward should be returned to the Blood Transfusion Laboratory if there is no prospect of it being transfused within 4 hours.
- If more than one unit of red cells are collected for a patient at one visit at a time, they should be transported in approved blood transport box.
- Blood components should not be transported between hospitals with the patient. Deviation from this policy should be reported immediately to the blood bank of the Royal Hospital.

Acceptance of blood products from a different hospital

Transport of blood components between hospitals should be always from a blood bank to a blood bank(according blood bank policy).

H. Identifying the patient immediately prior to commencing transfusion (bedside check)

- Any staff undertaking this procedure must be competent in the administration of intravenous drugs, have been trained in safe transfusion practice and have passed the required learning modules in accordance with the mandatory training and competence framework for blood transfusion.
- Blood transfusion must be checked by 2 trained staff independently for every blood component immediately prior to the transfusion as it supports and promotes the correct procedure, which includes key visual and verbal checks of patient identification and ensure an exact match between blood compatibility label and patient.
- Transfusions must not take place without a wristband attached and verified as correct for the patient which matches identically with the patient identification on the blood compatibility label.
- The 'begin transfusion' (bedside) check must always be performed at the patient's side immediately prior to starting of the transfusion. Once the check has been successfully completed, the blood component should be used to prime the blood giving set (if necessary) at the bedside and the transfusion started without delay. If the blood component left at the patient's side after the bedside check, or if another member of staff performed the check but did not start the transfusion, a repeat 'begin transfusion' (bedside) check will be required immediately prior to starting of the transfusion
- 2 member of staff, independently identified patient ID badge by using 3V/ bar code, is accountable for carrying out an identity check of the patient and the blood component at the patient's bedside. The person must be a healthcare practitioner who is currently registered.

back to the Blood bank to correct a mistake, depending on the cause of the mismatch. The incident must be reported in line with the Incident Reporting Policy.

- Positively identify the patient by asking his/her surname, first name and date of birth (whenever possible) and make sure that these patient identification details are the same as on the patient's wristband. It is essential that any patient having a blood transfusion has a wristband attached. If the wristband is removed, for example to take a blood test, the person removing it or finding that it has been removed is responsible for ensuring that a correct replacement wristband is attached immediately.
- For all patients, including unconscious patients, check that the following details(surname, first name, date of birth, medical record number and gender) are the same on:
 1. The patient's wristband
 2. The compatibility label attached to the blood component
 3. The prescription and / or medical records
- Check that the blood group and unit number on the blood bag are identical to those on the compatibility label attached to the blood bag

Also check that:

1. The blood has not passed its expiry date.
2. The blood bag shows no sign of damage and that there is no evidence of leakage
3. The blood component complies with any special requirements. If a patient has special requirements, such as irradiated blood components, the requirement must be noted on the patient's medical records and on the prescription.
4. The prescription is updated with the time and date of administration. Any transfusions taking place without the bedside transfusion checks must be fully documented, including blood unit number, recipient details and reasons for noncompliance, in the patient's health record and in response to follow-up documents which are sent in retrospect to ward managers in these circumstances.

Notes:-

Before the blood component is collected, the patient's vital signs (pulse, blood pressure, temperature and respirations) should be checked and recorded on the observation chart. This should not be done more than 30 minutes before collecting the blood from blood bank

Administration of blood and blood components:

- Before starting the transfusion, the patient must be positively identified in accordance with the processes set out in procedure using the 3V
- Transfusions must not take place without wrist band attached and verified as correct for the patient which matches identically with the patient identification on the blood compatibility label.
- The practitioner must wash his/her hands before starting the transfusion and utilize a no touch technique for the connection of the transfusion. Disposable, non-sterile gloves be worn.
- All blood components require the use of standard blood giving sets with appropriate filter (Please ensure that the right filter is used for the right component):
 1. Standard filter (170µm) or micro aggregate filter (40µm) must be used for red cells, FFP, cryoprecipitate and cryosupernatant, change every 12 hours if it continues blood transfusion.
 2. Platelets filter for platelets transfusions. One filter can cover 6 units of single donor platelets or one Apheresis unit.
 3. Filter should not be used for granulocyte transfusion
- Start the transfusion as soon as possible after the blood component's arrival in the clinical area. If a delay in starting the transfusion is likely, the blood should immediately be returned to the Blood bank refrigerator (within 30 minutes of removal) until just prior to the transfusion(please refer to page 12, Storage condition and Infusion Rate for Blood components).

The transfusion of each unit of red cells should normally be completed within one hour and a half and not more than **4 hours of removal of the blood from the Blood bank refrigerator.**

- Blood components must never be stored in drug or domestic refrigerators.
- If a unit of red cells has been out of the refrigerator for more than half an hour and there is no prospect of it being transfused, it must be returned to the Blood bank, explaining that it has been un-refrigerated for more than half an hour.
- Commence the transfusion adjusting the regulation clamp (or pump settings) to ensure the prescribed rate of blood flow.
- Medication must never be added to a unit of blood or infused in same IV line of blood.

- **Monitoring and Care of Patients Receiving Transfusion:**

The patient should be asked to report any potential adverse effects including shivering, rashes, flushing, and shortness of breath or pain in the extremities or joints.

Schedule of observations:

1. Record temperature, pulse, respiration rate and blood pressure before the start of each unit as part of the blood transfusion 'begin transfusion' (bedside) check

after 15 minutes from starting the transfusion, at the end of transfusion.

4. Record temperature, pulse, respiration rate and blood pressure at the end of each unit.

5. Note that signs of a severe transfusion reaction are most likely to become apparent during the first half an hour of transfusion of each component.

6. Changing the giving set in order to prevent bacterial growth, change the giving set and filter if the transfusion episode is to run for more than one unit

7. If a transfusion reaction is suspected, contact a doctor immediately, and record

8. Temperature, pulse, respiration rate and blood pressure. Further management depends on the type and severity of the reaction.

9. If a severe reaction is suspected:

- Stop the transfusion and seek urgent medical advice
- Change the giving set and maintain venous access
- The reaction MUST be reported to the Blood bank by submitting a filled transfusion reaction investigation form, the implicated unit and necessary blood samples.
- Record the volume and color of any urine passed. if there are signs of haemolysis the urine should be saved for analysis
- Observe the patient until haemodynamically stable, recording vital signs as per the policy of vital signs.

• **Documenting Blood Transfusions:**

- Each transfusion must be documented in the patient's medical records.
- The medical team responsible for the patient, must document in the patient's medical record the following information Consent to the transfusion and clinical indication for the transfusion. The nursing staff must fill completely the Transfusion Monitoring Chart on ALSHIFA system
- **Special considerations**

A. Transfusion and selection of blood components:

Before administering blood component, you should be familiar with the different types of blood components:

TRANSFUSING AND SELECTING BLOOD COMPONENTS

Blood component	Indications	Compatibility	Nursing considerations
• Red blood cells (RBCs) Same RBC mass as whole blood but with 80% of the plasma removed • Leukocyte-reduced RBCs Same as RBCs except with nearly 70% of the leukocytes removed	• Restoration or maintenance of oxygen-carrying capacity in symptomatic anemia • Increase RBC mass in acute anemia caused by trauma, surgical blood loss, or chemotherapy • Increase RBC mass in chronic anemia with associated cardiovascular decomposition • Avoid administration for anemia that's correctable with nutrition or drug therapy.	Group A receives A or O. • Group B receives B or O. • Group AB receives AB, A, B, or O (at discretion of the treating physicians). • Group O receives O. • Rh-negative recipients can receive only Rh-negative RBCs. • Rh-positive recipients can receive Rh-positive or Rh-negative RBCs.	• Transfuse RBCs through a sterile pyrogen-free transfusion set with an appropriate filter. • Prime the administration set with normal saline solution if using a Y-type administration set. • Begin the transfusion within 30 minutes of obtaining the blood from the blood bank. • Administer as the patient can tolerate, usually over 1½ to 2 hours, but complete the transfusion within 4 hours. • Know that RBCs can be separated into smaller volumes in special Paeds packs when longer infusion rates is required for patient at risk of fluid overload.

Platelets: • Single Donor Platelets units • Apheresis Units	• Bleeding caused by decreased circulating platelets or malfunctioning platelets • Increase platelet count in a patient who requires an invasive procedure • Prevent bleeding in a patient with a low platelet count	• Donor plasma should be ABO-compatible with the recipient's RBCs when large volumes are given to adults. • Rh-negative recipients should receive Rh-negative platelets, when possible, especially in women going through childbirth	• Transfuse through a sterile, pyrogen-free transfusion set with an appropriate filter. (Don't use a microaggregate filter.) A leukocyte-reduction filter may be necessary if leukocyte-reduced platelets aren't available. • Administration is commonly over 15 to 30 minutes Patients who don't benefit from transfusion of random donor platelets units may benefit from group specific Apheresis units or HLA-matched platelets.
Platelets: • Single Donor Platelets units • Apheresis Units	• Bleeding caused by decreased circulating platelets or malfunctioning platelets • Increase platelet count in a patient who requires an invasive procedure • Prevent bleeding in a patient with a low platelet count	• Donor plasma should be ABO-compatible with the recipient's RBCs when large volumes are given to adults. • Rh-negative recipients should receive Rh-negative platelets, when possible, especially in women going through childbirth	• Transfuse through a sterile, pyrogen-free transfusion set with an appropriate filter. (Don't use a microaggregate filter.) A leukocyte-reduction filter may be necessary if leukocyte-reduced platelets aren't available. • Administration is commonly over 15 to 30 minutes Patients who don't benefit from transfusion of random donor platelets units may benefit from group specific Apheresis units or HLA-matched platelets.
FFP: Noncellular portion of blood that's separated and frozen after donation and contains coagulation factors and other proteins	• Temporary warfarin reversal • Plasma exchange (especially for thrombotic thrombocytopenia) • Factor deficiency if concentrate is unavailable • Abnormal coagulation test results before invasive procedures • Liver disease with protein synthetic defect • Dilutional coagulopathy in massive transfusion cases • Consumptive coagulopathy	• Group A receives A or AB. • Group B receives B or AB. • Group AB receives AB. • Group O receives O, A, B, or AB. • Rh matching isn't required.	• Transfuse through a sterile, pyrogen-free transfusion set with an appropriate filter. • Infuse immediately after thawing. • Administration is usually over 30 to 60 minutes.

Cryoprecipitate no cellular blood component that's prepared by thawing fresh frozen plasma, recovering the insoluble precipitate, and then refreezing the precipitate within 1 hour of collection; contains concentrated levels of fibrinogen, factor VIII, von Will brand factor, factor XIII, and fibronectin	- Hypofibrinog enemia - Dysfibrinogenemia - Fibronectin deficiency	- ABO compatibility isn't required, but it is preferred. - Rh matching isn't required.	- Transfuse through a sterile, pyrogen-free transfusion set with an appropriate filter. - Infuse immediately after thawing. - It may be administered through a small-gauge IV catheter. - Administration is usually over 15 to 30 minutes.
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B. Storage condition and Infusion Rate for Blood components:

Red Blood Cells	Platelets	FFP/Cryoprecipitate
Storage condition: <u>In the Blood Bank:</u> - Refrigerated at 2-6C° <u>In the Ward:</u> - Never store in Domestic fridge. - Start infusion within 30minutes of removal from refrigerator at the blood bank	Storage condition: <u>In the Blood Bank:</u> 20-24C° on platelet agitator to maintain platelet and prevent clumping. <u>In the Ward:</u> 20-24C° - Should never be placed in a refrigerator. - Do not keep platelets on cold surfaces. - Must be transfused immediately.	Storage condition: <u>In the Blood Bank:</u> - Frozen> -20C° (Never ask for thawing unless you are sure and patient is ready for infusion; because once they are thawed they cannot be refrozen and will be discarded) <u>In the Ward:</u> 20-24C° - Must be transfused immediately.
Infusion Rate: 1. Adult: 90-120min/unit (Max 4. hrs); beyond 4 hours there is risk of bacterial contamination of units. 1. Paediatrics: 5ml/kg/hr*	Infusion Rate: 1. Adult Therapeutic dose: 30-60min. 2. Paediatrics Therapeutic dose: 10-20ml/kg/hr* If Platelet and other components collected at the same time, Infuse Platelet first.	Infusion Rate: 1. FFP: - Adult Therapeutic dose: 20ml/kg/hr. - Paediatrics Therapeutic dose: 10-20ml/kg/hr*. 1. Cryoprecipitate: - Therapeutic dose: 10-20ml/kg/hr.

All Blood components must be returned to blood bank within 30mins if they will not be transfused.

C. Blood Transfusion Reactions

A reaction to a transfusion may be mild or severe and life threatening. Adequate management depends upon the severity and likely nature of the reaction. It may be necessary to seek specialist advice from senior Haematology medical staff.

• Management of haemolytic transfusion reactions

ABO incompatible transfusions can cause hemolytic transfusion reaction. This reaction results from increased red cell destruction.. The reaction can be immediate or delayed.

Type of Reaction/ Description	Symptoms	Management
<u>Immediate Intravascular Haemolytic Reaction (usually ABO-incompatible Transfusions)</u> This may be caused by the transfusion of incompatible red cells, bacterially contaminated or thermally damaged blood. In a conscious patient only a few mls may be sufficient to cause a severe reaction which is seen within minutes of the transfusion. In an unconscious patient some of the symptoms will not be present. ABO incompatible transfusions usually occurs mainly due to clerical errors or bed side identification errors	<u>Symptoms (within 15-30 minutes)</u> • Fever, chills, rigor • Tachycardia • Hypotension and circulatory collapse • Severe pain at intravenous site • Pain in back or chest • Dyspnoea <u>Later symptoms</u> • Haemoglobinemia • Acute oliguria, renal failure and collapse Intravascular coagulation	• Stop the transfusion immediately • Resuscitate the patient and support him according to his vitals status • Send blood samples for: • CBC • Coagulation screen • LFTs • Haptoglobin • Blood culture • Renal function test • New crossmatch sample • Direct anti-human globulin (DAT) test • Return all blood units the Blood bank. Ensure that the giving set is removed before sending the unit. • Test all urine passed for haemoglobin • Maintain a strict fluid balance sheet • Continue observations
<u>Immediate Extravascular Haemolytic Reaction</u> Some antibodies to red cell antigens activate the complement pathway only weakly or not at all and destruction of the red cells takes place in the liver or spleen. This type of reaction is less severe	• Fever • Rigors • Jaundice (may occurs days later)	• Stop the transfusion immediately • Seek advice from Haematologist • Send blood samples for: • CBC • LFT • DAT • New cross match sample • Return blood units to Blood bank • Renal function test
Type of Reaction/ Description	Symptoms	Management
<u>Delayed Haemolysis</u> The titre of an antibody in a patient's plasma may be too low to detect in pre-transfusion tests. However if incompatible red cells are transfused a secondary response may be provoked which results in a rapid increase of antibody a few days later, thus causing the destruction of red cells	• Fever • Drop in haemoglobin level • Jaundice (often not before 5 days post transfusion, can be as late as day 10) • Haemoglobin (mean of 8 days post transfusion)	• Discuss with Haematology Consultant • Send blood samples for: • CBC • LFT • DAT • New cross match sample • Renal function test

<u>Reactions (FNHTR)</u> FNHTRs are unpleasant but not life threatening, but it should be remembered that a mild febrile reaction might be the early stages of an acute haemolytic reaction. FNHTRs are caused by cytokines in the donor or antibodies to the donor leucocyte antigens.	• Rise in temperature by 1°C from baseline. • Non-specific symptoms including chills and rigors. • Often symptoms seen towards the end of the transfusion	• Paracetamol is often all that is required • The rate of transfusion may be slowed • If the patient becomes unwell or hypotensive, the transfusion must not be restarted
<u>Allergic Reactions</u> Caused by antibodies to the infused plasma proteins or infusion of allergens that react with the patients IgE antibodies. It is more likely to occur with platelets and FFP.	• Urticaria • Itching	• Symptoms subside usually if transfusion is slowed • Give antihistamine, slowly by injection
<u>Anaphylaxis</u> This is a very rare but life threatening complication. The onset is rapid and often dramatic. In some cases it is associated with antibodies against IgA in a patient who has a severe IgA deficiency. Antibodies to other plasma proteins are implicated in other areas.	• Hypotension • Dyspnoea • Chest pain • Abdominal pain • Urticaria • Nausea and vomiting	• Resuscitate the patient and involve Medical High dependency physician from onset. • Immediate action required – discontinue transfusion • Maintain airway and give oxygen (40-100%) • Ensure that these medications in close proximity and consider giving them immediately: • Give adrenaline (0.5-1.0mg) intramuscularly • Repeat adrenaline every 10-20 minutes according to response in pulse rate • Give hydrocortisone (100-200mg) and chlorpheniramine (10-20mg over at least 5 minutes) – • both intravenously • Give nebulised salbutamol +/- aminophylline infusion • Do not restart transfusion • Track & Trigger observations • Haematology consult is recommended as well as notifying the blood bank.
<u>Infusion Of Bacteria Contaminated Blood components</u> Although this is extremely rare the mortality remains very high.	• Pyrexia $\geq 2^\circ\text{C}$ from baseline. • Rapid onset • Hypotension • Rigors • Collapse	• Discontinue transfusion • Immediate resuscitation with intravenous fluids and antibodies • Seek advice from Consultants for Haematology, Microbiology and Intensive Care Physician • Send blood samples for: • Blood cultures • Microbiological investigations of the remaining blood components • Return all blood units the Blood bank • Liaise with the bank to enable completion of appropriate documentation and notification of National Blood Service • Track & Trigger observations
<u>Transfusion Related Acute Lung Injury (TRALI)</u> This rare but life threatening complication is usually due to donor plasma that contains antibodies to the patient's leucocytes.	• Present as features of non-cardiogenic pulmonary oedema • Chills • Fever • Non-productive cough • Breathlessness	• Immediately seek advice from senior Haematology staff and ITU Physicians • Notify the National Blood Service, Trent Centre, Sheffield who will carry out appropriate further investigations • Consider transfer to an acute hospital • Resuscitate the patient

		• Haematologist must be involved • This reaction must be reported immediately to the blood bank • The Implicated unit must be returned to the blood bank • Send blood samples to rule out haemolytic transfusion reaction and bacterial contamination as mentioned above
<u>Fluid Overload</u> This can occur when correcting chronic anaemia in elderly patients or those with pre-existing cardiac disease	• Dyspnoea • Tachycardia • Hypotension	• Stop the transfusion • Give furosemide (40mg) intravenously • Arrange chest x-ray and ECG

• **Late Complications Of Transfusion:**

1. Iron overload

Transfusion dependent patients receiving red blood cells over a long period become overloaded with iron. Iron chelation therapy may be indicated in such cases.

2. Post-transfusion Purpura (PTP)

PTP is a rare but potentially lethal complication of red cell or platelet transfusions. It is often seen in female patients. PTP is caused by platelet-specific antibodies. It can typically be seen 5-9 days post transfusion, when the patient develops an extremely low platelet count with bleeding. Seek advice from the Consultant Haematologist.

References

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